Javad Vaez

Curriculum Vitae

mjvaez.cs@gmail.com
mjvaez
Google Scholar
Website

Professional Summary

Machine learning professional with a multidisciplinary background in AI, mathematics, and computer engineering. Hands-on experience developing a RAG prototype that enabled users to upload books and ask questions grounded in their content. Skilled in Python, machine learning, data analysis, LLMs, and research-driven problem solving. Currently pursuing a PhD at the University of Melbourne and conducting research in AI for biological and chemical data.

Technical Skills

Python, C/C++, MATLAB, LaTeX, HTML, CSS

Python Ecosystem	PyTorch, Scikit-learn, NumPy, Pandas, SciPy, Matplotlib, NetworkX, Jupyter
Software Engineering	Git, GitHub, Qt
Hardware Engineering	Hspice, VHDL, Proteus, AVR Microcontrollers

Education

2026 **PhD in Science**, *University of Melbourne*

- Member of the ARC Centre of Excellence for the Mathematical Analysis of Cellular Systems (MACSYS)

2022–2025 **M.Sc. in Computer Science (AI Specialty)**, *University of Tehran*, *GPA: 3.82/4*

- Ranked 1st in Iran overall by the QS and U.S. News rankings; in particular 1st in Computer Science and AI.
- **Thesis:** STAF: Sinusoidal Trainable Activation Functions for Implicit Neural Representation

2018–2022 **B.Sc. in Mathematics (Second Major)**, *University of Isfahan*, *GPA: 3.84/4*

2017–2022 **B.Sc. in Computer Engineering (Hardware Specialty)**, *University of Isfahan*, *GPA: 3.75/4*

Final Project: Applications of Group Theory and Automata Theory in Faster Searching of State Space

Professional Experience

2026-Present **Machine Learning Research Project**, *BioHub Cell Tracking During Development (Kaggle Competition)*

- Developing machine learning algorithms for automated cell detection, tracking, and lineage reconstruction from 3D time-lapse microscopy data.
- Investigating and developing graph-based optimization, motion modeling, and lineage inference for robust cell tracking.
- Designing evaluation and visualization tools to analyze tracking performance and guide model improvements.

2025-Present **English Language Tutor**

2025-2026 **Freelancer**
AI and Data Science

2021-2022 **Internship in Medical Equipment Development and Repair, *Shafa Medical Company***
Completed a hands-on internship at a company specializing in the manufacturing and repair of medical devices. Focused on hospital central sterilization systems, including autoclaves and plasma sterilizers, as well as respiratory support equipment like oxygen concentrators, BiPAP, and CPAP machines. Played a key role in programming and developing software for demo prototypes, facilitating rigorous testing and troubleshooting of new devices to ensure functionality and reliability.

Research Experience

2026 **A Unified Theory of Sinusoidal Activation Families for Implicit Neural Representations**, *Morsali, Alireza; **Vaez, MohammadJavad**; Soltani, Hossein; Kazerouni, Amirhossein; Taati, Babak; Mohammad-Noori, Morteza. Accepted by Transactions on Machine Learning Research (TMLR), 2026. OpenReview*

2023 **An alternative approach to inverse $\mathcal{Z}$-transform of rational functions**, ***Vaez, MohammadJavad**; Hosseini, Alireza; Jamshidi, Kamal. Journal of Engineering Mathematics, 151(1), 1–22 (2025). Published Version | Preprint | GitHub Repository*

2024-2025 **Analysis of the neurons of larval zebrafish**,
In collaboration with Dr. Dominic Burrows

2021–2023 **Random Generation of Group Elements Using Combinatorial Group Theory and Automata Theory, Along with a Hardware Example**[1], ***Vaez, MohammadJavad**; Kaedi, Marjan; Kalbasi, Mahdi. arXiv preprint arXiv:2311.16347 (2023). Preprint*

Honors and Awards

2025 **Admitted to the PhD programs in Electrical Engineering at the New Jersey Institute of Technology and Electrical & Computer Engineering at the Worcester Polytechnic Institute;** I was unable to pursue these opportunities due to the travel ban affecting Iranian students, link

2024 **Reviewer**, ICLR 2025 (International Conference on Learning Representations) – one of the highest-impact conferences in machine learning and AI research.

2022 **Second Rank among Math Students (Class of 2017)**, University of Isfahan.

2022 **Admitted to the Master's Program in Mathematics, Sharif University of Technology**, through direct admission based on academic excellence; declined the offer.

2020 $1.2 Reward Check from Donald Knuth for identifying a mistake in his book, *The Art of Computer Programming*[2]

Link of explanation

[1] Target journals include: *Combinatorics Probability and Computing*, *Random Structures and Algorithms*, and *International Journal of Algebra and Computation*.
[2] I included this in my CV because it was a unique and memorable reward.

Teaching Experience

Fall 2024 **TA for Advanced Theory of Algorithms**, University of Tehran, Instructor: Dr. Morteza Mohammad-Noori

This playlist includes presentations (in Persian) delivered by the students of this course.

Fall 2018 **TA for Calculus II**, University of Isfahan, Instructor: Dr. Ehsan Hakimian

Important Projects or Presentations

Bacterial Intelligence: How Bacteria Took the First Evolutionary Steps Towards Intelligenc: PDF

Manufacturing Workflow Management: GitHub Repository

Realtime Embedded Systems: LinkedIn Post

Buchberger Algorithm and Gröbner Basis in Inverse Kinematics of Manipulators: YouTube Video (Persian presentation)

Persian presentation of the paper "Learning the travelling salesperson problem requires rethinking generalization": YouTube Video

Presentation of the paper "BrainGB: A Benchmark for Brain Network Analysis with Graph Neural Networks": PDF File

Generally, this is my YouTube account on which I upload my presentations: youtube.com/@mohammadjavadvaez

This is my Stack Exchange account and you can see my activity: My Activity on Stack Exchange

Areas of Expertise or Interest

- Machine Learning, Neural Networks, Graph Neural Networks, Geometric Deep Learning
- Computational Neuroscience, Brain-inspired AI
- Signal Processing, Neural Signal Processing, Control Theory
- Reinforcement Learning (Certificates: Great Learning, LinkedIn Learning)
- Automata Theory, Finite State Machines, Cyber-Physical Systems, Digital Twins
- Hardware Engineering, VLSI, Neuromorphic Computing
- Mathematical Modeling, Stochastic Mathematics (including Randomized Algorithms, Probabilistic Graphical Models, Stochastic Cellular Automata, etc.)
- Bioinformatics, Computational and Mathematical Biology, Systems Biology, Biomedical Data Science

Creative Works

I compose Persian poems. You can read some of my works link

2015 Designing a tool to draw homographic functions, link

Languages

Persian Native

English Fluent, IELTS score: 7 (Speaking: 7.5)

Relevant A$^+$ and A Scores

Score	Courses
20/20	Artificial Intelligence & Expert Systems,Calculus II, Engineering Mathematics, Linear Control Systems, Fundamentals of Math. Analysis, Principles of Compiler Design, MATLAB Programming Workshop, Preliminary Number Theory
19.7/20	Statistical Methods
19.6/20	Local Differential Geometry
19.5/20	Fundamentals of Probability, Math. Analysis, Complex Functions, Very Large Scale Integration (VLSI), Computer Architecture Project, Digital Electronics Lab.
19.3/20	Presentation & Research Method, Data Transfer
19.25/20	Engineering Statistics & Probability, Calculus I, Fundamentals of Computer & Programming, Theory of Coding
19.06/20	Basic Physics II Lab.
19/20	Stochastic Processes, Probability I, Differential Equations, Algebra, Realtime Embedded Systems
18.5/20	Electrical Circuits, Physics II
18.3/20	Electronic Circuits
18.1/20	Advanced Theory of Algorithms
18/20	Partial Differential Equations (PDE), Manifold Geometry, Math. Software, Physics I, Electronic Circuits Lab.
17.85/20	Computer Aided Design Tools Lab.
17.78/20	Linear Optimization
17.75/20	Fundamentals of Economics
17.5/20	Advanced Computation Theory, Fundamentals of Algebra
17.4/20	Fundamentals of Computability Theory
17.3/20	Logic Circuits
17.25/20	Robotics
17.22/20	Interface Circuits Lab.
17.2/20	Image Processing
17.12/20	Algorithm Analysis & Design
17/20	Numerical Linear Algebra, Digital Electronics
16.75/20	Natural Language Processing
16.5/20	Microprocessors & Assembly Language, Computer Networks
16.1/20	Data Structures
16/20	Fundamentals of Matrix & Linear Algebra, Fundamentals of Math. Sciences

AI
Algorithms
Linear Algebra
Abstract Algebra
Computer Networks
Hardware Engineering
Mathematical Analysis
Geometry and Topology
Probability and Statistics
Theoretical Computer Science

Scores $\geq$ 18 are considered A$^+$, and scores between 16 and 17.99 are considered A.